



AI Transformation: Ready or Already?

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From the Editor

AI, anyone? A decade ago, the conversation was all about big data and machine learning. Today, the focus has shifted to generative models as a general-purpose technology. But what is actually required for successful adoption? Let me introduce three experts who have given this careful thought. Felix Dobslaw leads a new one-year master's program on AI transformation. He is joined by Lucas Gren, who drives this change in practice at the medtech company Getinge, and Erik Sterner, with whom he offers a popular free and open online course on practical generative AI. Together, we discuss AI transformation requirements across three dimensions: organizational, individual, and technological.—Markus Borg

IS GENERATIVE AI (GenAI) really transformational for the software sector? The signals point in opposite directions. On the one hand, we keep hearing that many AI initiatives fail—MIT Press has amplified that narrative.¹ On the other hand, some organizations report dramatic productivity gains alongside major workforce reductions. Klarna is a widely cited example, describing a shift from roughly 7,400 to about 3,000 employees in just two years while maintaining efficiency.² If both stories are true, *AI readiness* must be the real differentiator. In

this column, we discuss the requirements organizations must meet to realize the benefits of AI.

AI Transformation Readiness

Professional software development is, at its core, the art of building and working with useful abstractions. Over the last two decades, we've steadily moved from imperative programming—spelling out *how* to do something step by step—toward more declarative styles that focus on *what* we want. Concurrency handling is a good illustration. GenAI extends this trajectory by turning intent itself (that is, human-level instructions) into an interface. As a consequence, a much

broader group of people can now participate in building software.

Certain characteristics, capabilities, and markers seem to separate organizations that succeed from those that stall. In a race full of hype, teasing apart signals from noise is a shared challenge for researchers and practitioners alike.

The technology is moving so quickly that a cautious, evidence-first stance feels almost impossible. We have also changed our language, often talking about AI as if it were synonymous with GenAI and forgetting that the term *vibe coding* was coined only recently in February 2025. New large language models (LLMs) arrive monthly. Agentic architectures

evolve weekly. Interfaces change how people actually work—often faster than we can study them.

It feels a bit like the early COVID era, when medRxiv took off as labs rushed to share findings—except this time we’re not observing a primary and visible impact like an unfolding pandemic. Instead, we’re trying to understand a secondary one: what happens to organizations, work, and productivity when GenAI becomes broadly available.

And we’re not just changing the wheel while driving. We’re arguing about the destination. Are we trying to stay on the road and use AI to optimize existing processes, or are we rebuilding the entire vehicle—turning it into an airplane and attempting to takeoff in motion?

We believe the implications are immense, simply because of new ways humans can now communicate intent to machines. As the interface shifts, the transformation spans every area where software exists today, begging the question: are organizations ready for it? And what qualifies as ready?

AI transformation readiness is inherently sociotechnical. It succeeds only when the social system (*people* and their *organization*) coevolves with the *technology*. But readiness for what? Which parts of development do you intend to support with AI: coding, documentation, architectural analysis, or test case generation? This question is seldom spelled out, yet the answer shapes what readiness means and differs sharply by context. We structure the discussion around three dimensions that cut across these choices: organizational, human/individual, and technological.

Organizational Readiness

Organizational readiness starts with a genuine openness to change—even

radical change—and it must be visibly supported by executives and employees alike. It also requires clarity on what you want to achieve with AI assistance: which parts of development to support, and where the pay-off justifies the cost. That analysis should guide your choices, regardless of which solution provider you pick. From there, it becomes practical: people need room to experiment, along with low-friction access to the tools, data, and support that let them turn ideas into prototypes.

In large companies and high-reliability or heavily regulated domains, the usual structures often aren’t built for this. If every proof of concept is forced through production-grade requirements processes, governance gates, or legal reviews, experimentation stalls before it can demonstrate value. In those settings, readiness often means creating parallel innovation tracks—bounded sandboxes and pilots where teams can explore safely without being crushed by premature rigor.

Further, establishing a dedicated support function for GenAI at a sufficiently high level in the structure can, if done well, remove friction and speed up ideation across the whole organization.³ One key responsibility of such a function is to stay up to date with the already diverse and growing palette of tools, integrations, and models offered by GenAI providers.

It’s also easy to aim too low. Operational use cases are obvious (and an internal chatbot can be genuinely useful), but the transformation potential is often strategic: helping leaders synthesize signals, explore scenarios, and make decisions with more context and less delay. If GenAI is confined to the “efficiency” bucket, the organization

will miss its greatest leverage. Value creation is not fixed; with GenAI, companies can produce more customer value, so reducing ambition to “same value, but cheaper” misses most of the opportunity.

Finally, AI transformation readiness heavily depends on how well an organization understands itself—its roles, its processes, and how responsibilities and decision rights are delegated. The clearer the internal map is, the easier it becomes to see where AI can identify waste, augment work, and enable realistic automation. Just as importantly, it shows where AI would only create ambiguity or risk. That understanding also enables prioritizing AI investments and conducting a sober risk-benefit assessment before scaling pilots.

And while agility used to be a competitive advantage, GenAI turns it into table stakes. The pace of change is simply too fast for slow, linear adaptation. Staying competitive now requires the ability to learn, adjust, and redeploy capabilities continuously. As software becomes a commodity, flexibility must extend to software services and products themselves—for instance, in software-as-a-service dependencies, where solutions can be created in house quickly and tailored to local needs. There is, therefore, a greater need than ever for effective agile leadership across the organization.

Individual Readiness

Humans are at the center of all software change. Competency is central to individual readiness, especially the capacity to keep learning as roles and tools evolve. Upskilling and reskilling depend on it. In fast-moving software contexts, self-regulated learning becomes critical: setting

goals, seeking resources, and adapting without formal training.

How people feel about AI and how willing they are to engage with it also shape readiness. Transformation follows what people want and how they collaborate in practice. It is therefore essential to meet employees where they are to understand how they think and how they relate to new technology and its implications.

Last fall, we introduced a free, open online course on working and learning with GenAI.⁴ To understand individual readiness requirements, we asked participants at the

start of the course a simple question: “What are your thoughts and feelings about AI development in professional practice?”

More than a thousand people registered for the course, and 328 answered our survey. Participants were mainly from Europe (70%) and the United States (15%). Approximately half identified as male and half as female. Most respondents worked in industry (about two-thirds), while roughly 10% were from academia and slightly fewer from the public sector. About half worked in management and administration, just over a quarter in engineering, R&D,

and IT/data roles, and the remainder across other professions.

Figure 1 summarizes the results. Overall, 59% of respondents expressed uncertainty or negative feelings toward GenAI. Notably, this sample comes from a population that voluntarily signed up for a 25 h GenAI course. If anything, we would guess this group to be more positive than the broader workforce—but that remains to be investigated.

If uncertainty and negative sentiment are this prevalent among motivated participants, individual AI readiness clearly depends on addressing these concerns. Training programs play an important role, but so does transparent and continuous communication about AI strategy within the organization. This includes how the organization interprets ongoing developments in AI as well as openly stating what is unknown.

Technological Readiness

AI is not a quick fix for something that is broken. On the contrary, solid engineering practices are what make an organization technologically ready for AI transformation. Existing quality assurance processes are your safety net and foundation. As the cost of adding code approaches zero, organizations can quickly spiral into the deepest technical debt they have ever seen. We bet that many reckless projects will confront existential risks.

When introducing AI, rapid feedback loops on new or modified code are what save you from steering off the road. Once an initiative is underway, continuous feedback and follow-up at the program level let teams adjust scope and ambition as they learn—as with any innovation effort. Basic building blocks, such



FIGURE 1. Sentiment analysis of responses (n = 328) to the question: “What are your thoughts and feelings about AI development in professional practice?”

as configuration management, continuous integration/deployment, and high-quality test suites, must be the backbone of the transformation. If you're not there yet, return to step one. Experience reports increasingly show that AI assistance tends to amplify existing weaknesses rather than compensate for them.

Among these building blocks, testing deserves special mention. While the others are more binary in nature, testing quality lies on a spectrum. Broad AI adoption requires efficient and effective test suites. They must provide good coverage, and they'll be run more often than ever before. Don't let AI loose unless you can verify that the results align with your goals.⁵ Specifying requirements as test cases and test-driven development get another boost now. Also, test that your tests are meaningful—poorly aligned AI can unfortunately remove test cases just to make the suite pass.

Another consideration relates to the quality of the pre-AI code itself. As with human developers, LLMs perform better on simple, easy-to-read code. Using refactoring as a proxy for development activities, our recent research shows that AI is far less likely to break functionality during refactoring when code health is high.⁶ In other words, our new virtual coding buddies also get lost in complex code. When planning an AI rollout, we recommend starting with pilot projects in parts of the codebase that are “AI-friendly.”

Are You Ready for This?

How do you know whether you meet the requirements discussed so far? Here, we outline four building blocks that we consider critical for a successful transformation. In “Zooming Out—Is the World Ready?,” we take a broader perspective.



ZOOMING OUT—IS THE WORLD READY?

On a societal level, we still need to grapple with the challenge of timely and lasting education. Upskilling and reskilling must help individuals orient themselves and function in a rapidly changing AI-infused labor market, including coding professions. Coping with the pace of change requires new perspectives on education that emphasize self-regulation, adaptability, and resilience. Probably even more so for software professionals. We may well be experiencing a technology explosion in which the coming ten years bring as much development as the last hundred.^{S1}

There is reason for optimism. How to feasibly assure high software quality remains a largely unsolved problem, and AI might help address it. At the same time, the long-discussed global undersupply of software professionals appears to be easing, which gives hope in a world that depends on software as critical infrastructure. Yet the path forward isn't clearly sustainable, economically or environmentally. Sensible business models must account for the growing reliance on compute infrastructure and its carbon footprint. Winning the AI race cannot come at the cost of a crashing society.

Reference

S1. W. MacAskill and F. Moorhouse, “Preparing for the intelligence explosion,” 2025, [arXiv:2506.14863](https://arxiv.org/abs/2506.14863).

1. *Obtain a clear understanding of the needs and bottlenecks in the part of the organization to be transformed.* The potential for value creation with GenAI is maximized when related organizational processes, roles, and responsibilities are well understood. Start by examining constraints on a case-by-case basis and focus on simpler ones with favorable effort-outcome ratios.⁷
2. *Nurture employees' autodidactic abilities.* Organizations need to take responsibility for creating environments that support continuous learning and exchange and foster a culture of knowledge sharing and experimentation that encourages initiative.
3. *Make space for internal R&D.* Relying solely on external advice means lagging behind. Change is moving too fast for that. Organizations must invest internally in structures for knowledge exchange, systematic experimentation, and active participation in the broader AI debate.
4. *Organizations need technical mechanisms to continuously assess risk, observe AI behavior, and log outcomes over time.* As agent autonomy increases, this becomes critical. Deployment will never be a one-off event. Logging must be continuous and deliberate. Tooling



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and infrastructure are needed to make guardrails and quality gates scalable, for example, model context protocol servers for the agents, code analysis in the continuous integration pipeline, and human oversight where it pays off.

who understand the potential of AI for their specific needs and succeed in embracing it in inclusive and empathetic ways will be better equipped to navigate competition and inevitable hurdles along the winding road ahead.

We'd be happy to continue the discussion. What are your thoughts? Where is the world heading? Please reach out and share your perspective! 📧

References

1. A. Challapally, C. Pease, R. Raskar, and P. Chari, "The GenAI divide:

State of AI in business 2025," MIT Nanda, Cambridge, MA, USA, pp. 1–75, 2025. [Online]. Available: https://mlq.ai/media/quarterly_decks/v0.1_State_of_AI_in_Business_2025_Report.pdf

2. P. Fore, "AI enabled Klarna to halve its workforce—Now, the CEO is warning workers that other 'tech bros' are sugarcoating just how badly it's about to impact jobs," *Fortune*, Oct. 25, 2025. [Online]. Available: <https://fortune.com/2025/10/10/klarna-ceo-sebastian-siemiatkowski-halved-workforce-says-tech-ceos-sugarcoating-ai-impact-on-jobs-mass-unemployment-warning/>
3. L. Gren and R. Feldt, "Cross-functional AI Task Forces (X-FAITS) for AI transformation of software organizations," in *Proc. 29th Int. Conf. Eval. Assessment Softw. Eng.*, 2025, pp. 793–796, doi: [10.1145/3756681.3757015](https://doi.org/10.1145/3756681.3757015).
4. "Open online courses | Chalmers." Chalmers tekniska högskola. Accessed: Feb. 14, 2026. [Online]. Available: <https://www.chalmers.se/en/education/programmes-and-courses/continuing-professional-development/open-online-courses/working-and-learning-with-generative-ai/>
5. L. Gren and F. Dobslaw, "The expert validation framework (EVF): Enabling domain expert control in AI engineering," in *Proc. 5th Int. Conf. AI Eng. – Softw. Eng. AI*, 2026.
6. H. Borg and S. Tornhill, "Code for machines, not just humans: Quantifying AI-friendliness with code health metrics," in *Proc. 3rd Int. Conf. AI Found. Models Softw. Eng.*, 2026.
7. S. Wang, Y. Yu, R. Feldt, and D. Parthasarathy, "Automating a complete software test process using LLMs: An automotive case study," in *Proc. 47th Int. Conf. Softw. Eng.*, 2025, pp. 747–747.

Organizations around the world are figuring out how to take a pole position in this AI transformation race. Those